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Saveetha Institute of Medical And Technical Sciences

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TECHNICAL PRESENTATION

Big Data Analytics in Cloud Computing

Team No. [12]

CSA15 – Cloud Computing and Big Data Analytics

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Abstract

- **Big data:** huge volume, velocity, variety
- **Problem:** on-premise systems are costly and hard to scale
- **Cloud:** elastic storage, compute, managed analytics
- **Architecture:** data lake, batch + stream, warehouse
- **Platforms:** AWS, Azure, Google Cloud
- **Value:** faster insight, lower cost, better decisions
- **Case study:** Netflix on AWS

1 Architectures
Cloud data-processing layers

2 Platforms
AWS • Azure • Google Cloud

3 Value & Use Cases
Business impact, Netflix case

Aim and Objective

AIM

Understand how cloud computing enables scalable, cost-effective big data analytics

1. **Explore architectures**
   Cloud data-processing layers and pipelines
2. **Compare platforms**
   AWS, Azure and Google Cloud analytics services
3. **Assess business value**
   Cost, speed, scale and decision quality
4. **Study real applications**
   Netflix case study on AWS

Introduction

- **Big data:** datasets too large or fast for traditional tools
- **Sources:** IoT, social media, logs, transactions
- **Cloud computing:** on-demand compute and storage [3]
- **Pay-as-you-go,** elastic, globally available
- **Cloud + big data:** analytics without owning hardware

The 5 Vs of Big Data

Volume
TB to petabyte scale

Velocity
Real-time data streams

Variety
Structured to unstructured

Veracity
Data quality and trust

Value
Insight that drives decisions

Core Technical Explanation

PROBLEM STATEMENT

Traditional on-premise analytics cannot economically store and process growing, fast, diverse data

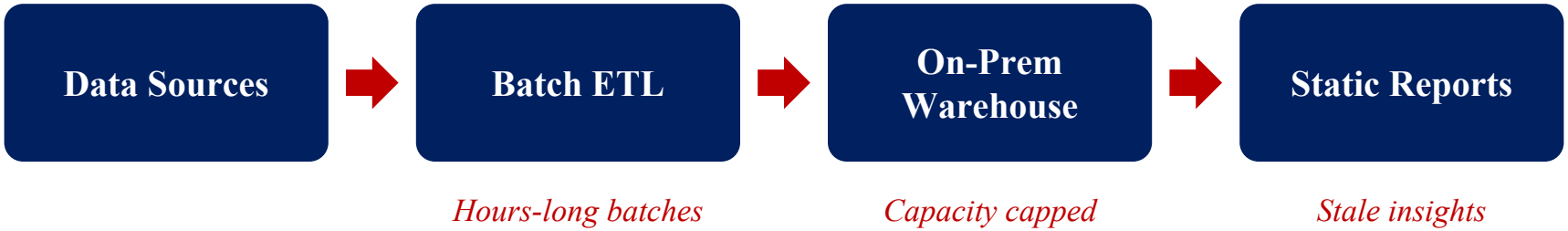
Scalability	Cost	Latency	Governance
Fixed hardware, long procurement, peak overload	High up-front CapEx, idle capacity most of the time	Batch-only jobs deliver late insights	Data silos, security and compliance gaps

Research need: *elastic, secure, unified analytics on the cloud [1]*

Existing System

- **On-premise** Hadoop clusters and enterprise data warehouses
- **Fixed capacity** bought up-front (high CapEx)
- **Batch ETL:** insights arrive hours or days late
- **Hard to scale** for peaks; idle in quiet periods
- **Siloed data** and heavy manual maintenance

Typical on-premise pipeline



Proposed System

Cloud-native analytics platform: managed, elastic, pay-per-use

1 Elastic Storage

Data lake on object storage; scales to petabytes

2 Managed Processing

Batch and stream engines without cluster upkeep

3 Serverless Warehouse

SQL analytics; pay only for queries run

4 Self-Service BI & ML

Dashboards, predictions and AI services

5 Security & Governance

Encryption, access control, audit trails

6 Pay-as-You-Go

Scale up for peaks, scale down to save

Overall System Architectural Design

Data Sources	Ingestion	Storage	Processing	Analytics
IoT & sensors	Batch upload	Data lake (S3, ADLS, GCS)	Batch: Spark / EMR	Warehouse (BigQuery, Redshift)
Apps, logs, clickstream	Streaming (Kafka, Kinesis)	Raw → curated zones	Stream: Flink, Dataflow	BI dashboards
Social, web, databases	APIs & CDC	Lakehouse (Delta, Iceberg)	ETL: Glue, Data Factory	ML & AI models

Security • Governance • Monitoring • Orchestration

Layered, loosely coupled design: every layer scales independently

Proposed Methodology and Algorithm

Analytics pipeline: raw data to decision



Algorithm: MapReduce [4] on Apache Spark [5]



- **Parallel:** workers process data blocks simultaneously
- **Fault tolerant:** failed tasks re-run automatically
- **Elastic:** add nodes on demand, release when done

Cloud Analytics Platforms

Layer	AWS	Microsoft Azure	Google Cloud
Storage	Amazon S3	ADLS Gen2	Cloud Storage
Batch / ETL	EMR, Glue	Databricks, Data Factory	Dataproc, Dataflow
Streaming	Kinesis, MSK	Event Hubs	Pub/Sub, Dataflow
Warehouse	Redshift	Synapse Analytics	BigQuery
BI & ML	QuickSight, SageMaker	Power BI, Azure ML	Looker, Vertex AI

Choice depends on existing stack, skills, pricing and compliance needs

Business Value

Lower Cost

- No up-front hardware (CapEx to OpEx)
- Pay only for what you use

Faster Insights

- Real-time streams, self-service BI
- Shorter time from data to decision

Scale & Agility

- Petabytes on demand, global reach
- Launch new analytics quickly

Smarter Decisions

- Personalization, fraud detection, forecasting
- ML and AI services built in

ROI = faster decisions + lower infrastructure cost + data-driven products

Real-world Use Case: Netflix on AWS

Netflix Keystone: Kafka-based event pipeline on AWS [2]

~500 B	~1.3 PB	~8 M
events per day (2016)	data per day	events per second at peak

How the data flows



- **Insights drive** recommendations, playback quality and operations
- **Volume has grown** past 1 trillion events per day since 2016

Conclusion

Problem	Solution	Impact
• Legacy analytics: costly, rigid, slow • Data growing in volume, speed, variety	• Cloud data lake with batch + stream processing • Managed platforms: AWS, Azure, GCP	• Elastic scale and lower cost • Faster, data-driven decisions • Proven at Netflix scale

VALUE & SIGNIFICANCE

Cloud makes big data analytics scalable, affordable and accessible

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[4] J. Dean and S. Ghemawat, "MapReduce: Simplified Data Processing on Large Clusters," Communications of the ACM, vol. 51, no. 1, pp. 107–113, 2008.

[5] M. Zaharia et al., "Apache Spark: A Unified Engine for Big Data Processing," Communications of the ACM, vol. 59, no. 11, pp. 56–65, 2016.

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